

Assignment 1

Due date: Sunday, October 11, at 11:59 PM. Late assignments will not be accepted.

You can work on this assignment in groups. However, everybody must hand in their own answers. I recommend that you type your answers in LaTeX (figures and long derivations can be done by hand). Assignments are to be submitted to the Dropbox in .pdf format. There are several scanning apps that allow you to scan and convert handwritten notes to .pdf.

Question 1: Consider an agent who participates in an economy with only $L = 2$ commodities. Assume that the choice function $x(p, w)$ is single-valued and that the agent's choice behavior satisfies Walras' law and the weak axiom of revealed preferences (WARP).

1.1 Using only what you know from Chapter 2, prove that the Slutsky matrix is symmetric [Hint: you can either combine Propositions 2.E.1 and 2.E.2 or use Proposition 2.F.3]. Explain how you made use of the various assumptions.

1.2 Does your proof from 1.1 extend to the case where $L = 3$? If yes, explain why. If no, explain why not.

1.3 In this part, you may use only what you know from Chapter 2 and what you learned from 1.1 and 1.2, along with the definition of SARP. Do you agree with the statement: "SARP implies that the Slutsky matrix is symmetric and negative semidefinite when $L = 2$ "? Does your argument stand when $L = 3$? Explain your answers.

Question 2: The following table summarizes the data that was gathered by following a consumer for 3 years. Bold entries indicate which bundle was chosen in which year. The entries in the table are the amount of money required to purchase the bundle in question at the prices in question. Assume that Walras' law is satisfied.

	$p^1 = (1, 2)$	$p^2 = (1, 1)$	$p^3 = (1, 1)$
$x^1 = (12, 3)$	18	15	15
$x^2 = (5, 5)$	15	10	10
$x^3 = (3, 12)$	27	15	15

2.1 Which bundles have been revealed preferred to which? Is WARP satisfied? Prove and explain your answers.

2.2 Assume for this part that the choices in the table were made by a consumer with a rational preference relation. Assuming preferences are convex and monotonic, is $x^4 = (4, 10)$ better or worse than x^2 and x^3 ? Is it possible to do the comparison assuming only convexity? Prove and explain your answers [it may be helpful to include a figure].

2.3 If x^4 is observed in year 4 at prices $p^4 = (2, 1)$, which bundles are revealed preferred to which? Is WARP satisfied? Explain your answers.

Question 3: Consider a consumer whose preferences can be represented by the following utility function:

$$u(x_1, x_2) = (x_1 - 1)(x_2 - 1).$$

3.1 Sketch the indifference set where utility is 0. Indicate the superior set (be careful!). In a new figure, sketch the indifference set where utility is 1.

3.2 Are the preferences convex? Are the preferences locally non-satiated? Are they monotone? Explain your answers.

3.3 Consider the following utility function instead:

$$v(x_1, x_2) = \begin{cases} (x_1 - 1)(x_2 - 1) & \text{if } x \gg 1 \\ 0 & \text{otherwise} \end{cases}$$

Sketch the indifference set where utility is 0, and indicate the superior set (be careful!). Sketch the indifference set where utility is 1.

3.4 Are the preferences that are represented by $v(x_1, x_2)$ convex? Are they locally non-satiated? Are they monotone?

3.5 Assuming $p_1 + p_2 < w$ (the bundle $(1, 1)$ is affordable), do you agree that the same bundle is optimal for both utility functions? Explain your answer.

Question 4: Consider a consumer whose preferences, $\succsim$, on the consumption set $X = \mathbb{R}_+^n$ can be represented by an *additively separable* utility function of the form

$$u(x_1, x_2, \dots, x_n) = \sum_{i=1}^n f_i(x_i), \quad (1)$$

where each $f_i(x_i)$ is a strictly concave function (not necessarily differentiable) in the single variable x_i , $i = 1, 2, \dots, n$.

4.1 Are the preferences represented by (1) strictly convex? Prove and explain your answers.

4.2 Does your answer to 4.1 change if $f_1(x_1) = x_1$ (while still assuming that all the other $f_i(x_i)$ are strictly concave in their argument)? What if $f_1(x_1) = x_1$ and $f_2(x_2) = x_2$? Prove and explain your answers.